

Recognizing Stress Using Semantics and Modulation of Speech and Gestures

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Abstract—This paper investigates how speech and gestures convey stress, and how they can be used for automatic stress recognition. As a first step, we look into how humans use speech and gestures to convey stress. In particular, for both speech and gestures, we distinguish between stress conveyed by the intended semantic message (e.g. spoken words for speech, symbolic meaning for gestures), and stress conveyed by the modulation of either speech and gestures (e.g. intonation for speech, speed and rhythm for gestures). As a second step, we use this decomposition of stress as an approach for automatic stress prediction. The considered components provide an intermediate representation with intrinsic meaning, which helps bridging the semantic gap between the low level sensor representation and the high level context sensitive interpretation of behavior. Our experiments are run on an audiovisual dataset with service-desk interactions. The final goal is having a surveillance system that would notify when the stress level is high and extra assistance is needed. We find that speech modulation is the best performing intermediate level variable for automatic stress prediction. Using gestures increases the performance and is mostly beneficial when speech is lacking. The two-stage approach with intermediate variables performs better than baseline feature level or decision level fusion.

Index Terms—Stress, surveillance, speech, gestures, multimodal fusion, semantics, modulation, affective computing, stress recognition

1 INTRODUCTION

WHILE automatic detection of unwanted behavior is desirable and many researchers have delved into it, there are still many unsolved problems that prevent intelligent surveillance systems to be installed to assist human operators. One of the main challenges is the complexity of human behaviour and the large variability of manifestations which should be taken into consideration.

Emotions and stress play an important role in the development of unwanted behavior. In [1] it is distinguished between instrumental and affective aggression. Instrumental aggression is goal directed and planned, e.g. pick-pocketing. Affective aggression results from strong emotional feelings, like anger, fear and frustration. Furthermore, there is a link between stress and aggression [2], [3]. Detecting negative emotions and stress at an early stage can help preventing aggression. However, an early stage is characterized by more subtle behavior compared to violence, which increases the difficulty of automatic detection.

Stress is a phenomenon that causes many changes in the human body and in the way in which people interact. It is a psychological state formed as response to a perceived threat, task demand or other stressors, and is accompanied by specific emotions like frustration, fear, anger and anxiety. For a thorough overview of stress we refer the reader to [4].

Since the final application of this work is in the surveillance domain, we are interested in the stress observed in the overall scene, and not per person.

We consider the case of supervising human-human interactions at service desks based on audio-visual data. However, the service desk domain is considered as a proof of concept. Human operated service desks are places where cases of urgency, overload and communication problems are likely to occur. The situation is similar for virtual agent systems employed in public service, where stress can arise due to task complexity and the inability of the virtual agent to act as expected by the client. Our goal is to automatically detect when stress is increasing and extra assistance is needed.

People use a variety of communicative acts to express semantic messages and emotion. Speech is used to communicate via the meaning of words, as well as via the manner of speaking. Several other nonverbal cues like facial expressions, gestures, postures and other body language are used in communication. We are interested in how these verbal and nonverbal cues are used in conveying stress, and how they can be used to automatically assess stress. In particular, our attention focuses on speech and hand gestures (hereinafter called gestures), since they are rich sources of communication and promising for automatic assessment.

While speech is generally considered the primary means of communication, [5] and [6] emphasize the importance of gestures. However, contradicting findings are presented in [7], where it is suggested that gestures have no additional communicative function compared to speech. Motivated by these contradictions on the communicative function of gestures in general, we study the role of gestures in assessing stress.

For automatic behavior analysis, typically low level features are mapped to a ground truth. A known problem with this approach is the semantic gap between the high level

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context sensitive interpretation of behavior and the low level machine generated features. We expect that this problem is likely to occur in the case of automatic stress prediction, since stress is a complex concept and has a large variety of manifestation. A possible solution for bridging the semantic gap is to consider a decomposition of stress into variables that might be easier to predict, and build a final decision based on them.

To summarize, we formulate our research questions as follows:

- 1) What is the contribution of verbal and nonverbal communicative acts in conveying stress?
- 2) Do gestures contribute to communicating stress and automatic stress assessment, or is all the information already included in speech?
- 3) Which intermediate features can be used in the framework of automatically assessing stress, what are their unique predictive values and which are good combinations that complement each other?
- 4) What is the impact of using intermediate level features compared to low level sensor features only?

To research these questions we use a dataset of audio-visual recordings of human-human interactions at service desks. The recordings contain a variety of scenarios in which stressful situations arise. The scenarios were freely improvised by actors, which means that the interaction was building up naturally, by spontaneously reacting to one another.

Our first step is to study how humans perceive stress from the recordings, based on cues from speech and gestures. We propose a human model for conveying and perceiving stress by speech and gestures, and based on annotations, we analyze which are the dominant communicative acts in conveying stress. As a second step, we propose a model for automatic stress assessment. It is based on a three level architecture, and the intermediate level is inspired by the variables proposed in the human model.

This paper is organized as follows. In Section 2 we give an overview of relevant related work. Next, in Section 3, we present our models for conveying and automatically assessing stress which we use for answering the research questions. We continue in Section 4 with descriptions of the dataset and its annotations. Section 5 focuses on the intermediate level variables proposed for the automatic stress assessment framework. Next, we give details on the experimental setup in Section 6, including a description of the low level features used for automatic stress assessment, the classification procedure and segmentations. Details on how stress is conveyed by speech and gestures based on the human model are presented in Section 7. We present and compare our automatic stress prediction method to a baseline stress prediction from low level features in Section 8. The paper ends with a summary and conclusions in Section 9.

2 RELATED WORK

This section highlights relevant related studies, ranging from identifying stress cues in speech and gestures, to automatic behavior recognition based on speech, gestures and multi-modal data. We end with outlining works that use

intermediate representations and explain what makes our approach different.

Many studies investigate clues in verbal and nonverbal communication, and how they are used for communication and affective displays. A comprehensive set of acoustic cues, their perceived correlates, definitions and acoustic measurements in vocal affect expression is provided in [8]. The same work also provides guidelines for choosing a minimum set of features that is bound to provide emotion discrimination capabilities. An extensive study [9] presents an overview of empirically identified major effects of emotion on vocal expressions. In [10] more emphasis has been put on voice and stress. The most important acoustic and linguistic features characteristic for emotional states from a corpus of children interacting with a pet robot are identified in [11]. In [12] and [13], different categories of nonverbal behavior are identified, part of them having the function of communicating semantic messages and part of them of transmitting affect information. These investigations point to the suitability of considering speech and gestures for assessing stress.

The research described in [14] focuses on analyzing, modeling and recognizing stress from voice, using recording of simulated and actual stress [15]. The study in [16] considers discriminating normal from stressed calls received at a call-center. Multiple prediction methods applied to different feature types are fused, resulting in a significant improvement over each single prediction method. The work of [17] focuses on automatic aggression detection based on speech.

The relationship between emotion and hand gestures (more specifically handedness, hand shape, palm orientation and motion direction) has been investigated in [18]. The study concluded that there is a strong relation between the choice of hand and emotion. An interesting approach for recognizing emotion from gestures is presented in [19]. Instead of trying to recognize different gesture shapes that express emotion, the authors use nonpropositional movement qualities (e.g. amplitude, speed and fluidity of movement). Most relevant for our work is the approach in [19], which proves that emotion recognition from gestures can be performed without specifically recognizing the gestures, but rather focusing on how they are done.

The added value of using multimodal information instead of unimodal has been highlighted in previous studies. Research in [20] addresses multimodal emotion recognition using speech, facial expressions and gestures. Their algorithms were trained on acted data for which actors were supposed to show a specific gesture for each emotion. Fusing the three types of features at both feature and decision level improved over the best performing unimodal recognizer. Fusion of cues in the face and in gestures have been used for emotion recognition in [21]. Again, a gesture type has been considered for each emotion. For surveys on multimodal emotion recognition we refer the reader to [22] and [23]. Future directions outlined in [23] include developing methods for spontaneous affective behavior analysis, which are addressed in this paper.

Several studies reflect the benefits of using intermediate representations in order to automatically assess a final concept. In [24], the focus is on detecting violent scenes in videos for protection of sensitive social groups by audio-visual data analysis. A set of special classes are detected, like

music, speech, shots, fights and screams and the amount of activity in video. From video, the amount of activity in the scene was used to discriminate between inactivity, activities without erratic motion and activities with erratic motion (fighting, falling). A two stage approach for detecting fights based on acoustic and optical sensor data in urban environments is presented in [25]. Low level features are used to recognize a set of intermediate events like crowd activity, low sounds and high sounds. The events based on video data are related to the behavior of crowds: normal activity, intensive activity by a few or by many persons, small crowd or large crowd, and also it is distinguished between different categories of sounds. The highest performance is achieved by fusing the cues from both modalities, again pointing that multimodal fusion is an interesting direction to explore for stress prediction. In the work of Grimm et al. [26], the continuous representation of emotions composed of valence, activation and dominance was used as an intermediate step for classifying emotion categories.

Interesting studies can be found in literature both with respect to automatic emotion assessment as well as automatic surveillance. We see our work to be at the intersection between these two fields. What makes our research different is that we take inspiration from the human model of stress perception adding intermediate level variables based on how and what speech and gestures communicate stress. The considered case study is challenging due to the nature of the data. Unlike in the mentioned emotion recognition studies, we do not have a specific gesture type that appears per emotion. A large variety of gestures appear spontaneously but also many times there are no gestures or they are not visible, and speech is spontaneous and sometimes overlapping. Our contribution is a novel two-stage stress prediction model from low level features using an intermediate level representation of gestures and speech.

3 METHODOLOGY

In this section we present the methodology for our research. Since our final goal is to develop an automatic system that is able to assess stress, we regard stress from two perspectives. The first perspective is the one of human expression and perception of stress. In this case the focus on how people convey and perceive stress and what communication channels they use, with a focus on speech and gestures. The second perspective is the one of an automatic stress recognition system based on sound and vision, and available low level features. The components of our model for automatic stress assessment are inspired from the human model. Our analysis is performed from the perspective of automatic surveillance. In that sense, by stress we refer to the global stress perceived in the scene and not to the stress of one person. In the study based on the human model as well as in the study related to automatic stress assessment, we use footage from one audio-visual camera and consider all audible speech and all visible gestures.

3.1 Model of Stress Expression and Perception Using Speech and Gestures from a Human Perspective

The model we present in this section is a basic model that is not meant to be complete or novel. Rather, it has the goal

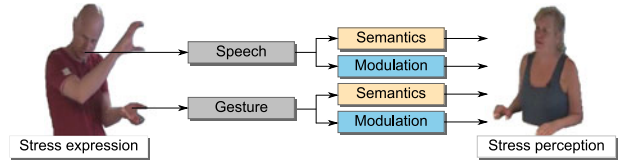


Fig. 1. Human model of conveying and perceiving stress by the components of speech and gestures.

and function of enabling us to operationalize phenomena about stress that relate to gestures and speech.

Following [27] and [28], we use the term *semantics* to denote the information that contributes to the utterance's intended meaning. Following [29], we use the term *modulation* to denote the part of the message that was transmitted by how the semantic message was delivered.

For both speech and gestures, we observe that there are two ways in which they can communicate stress, which refer to *what* is being communicated (the semantic message) and *how* the message is being communicated (modulation). For example, speech can convey stress by the meaning of the spoken words (e.g. "I am nearly missing the flight."), but also by the way in which the utterance was voiced (e.g. someone speaks loud, fast, with a high pitch). In analogy to speech, gestures can also communicate stress in two ways: by the common meaning of the gesture - *what* the gesture is saying (e.g. a pointing to self sign), and by *how* the gesture is being done (e.g. a gesture is sudden, tense, repetitive).

As illustrated in Fig. 1, the proposed model consists of four components:

- *Speech semantics stress (SSS)*. The extent to which stress is conveyed by the spoken words (the linguistic component).
- *Speech modulation stress (SMS)*. The extent to which stress is conveyed by the way the message was spoken (the paralinguistic component of conveying stress).
- *Gesture semantics stress (GSS)*. The extent to which a gesture is conveying stress based on its meaning i.e. the interpretation of the sign.
- *Gesture modulation stress (GMS)*. The extent to which a gesture is conveying stress based on the way it is done, i.e rhythm, speed, jerk, expansion, and tension of the gesture.

We are interested in how stress is conveyed by these four means of communication. In Section 7 we explore if any of them is dominant, how they are correlated and how well their annotated labels perform in predicting stress.

3.2 Model for Automatic Stress Assessment Using Intermediate Level Variables

As defined in the introduction, we are interested in which components of speech and gestures are good clues when it comes to assessing stress. Starting from the human model of perceiving stress as introduced in Section 3.1, we propose a three level architecture for automatic stress assessment.

The automatic stress assessment model is depicted in Fig. 2. The low level consists of sensor features, the intermediate level of variables related to stress inspired from the human model, and the last level of the final stress assessment.

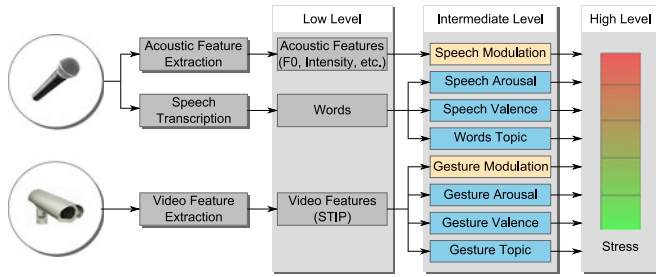


Fig. 2. Automatic model for assessing stress using low level features and intermediate level variables.

By using a classifier and the labels for the stress variable, we can compute the relation between the low level acoustic and video features and the high level stress variable. This enables us to automatically compute the stress level based on low level features. We refer to this method as the *baseline* model.

A drawback of the baseline model is the semantic gap between the low level features and the high level interpretation. To improve the baseline automated model, we defined variables on the intermediate level. There is a difference between the acoustic and video features from the low level and the variables in the intermediate level. The acoustic and video features are computed automatically but they have no intrinsic meaning. The intermediate level variables are automatically computed as well, yet in contrast to the low level feature they do have a meaning with respect to stress.

A mapping between the human model variables in Fig. 1 and the intermediate level variables in Fig. 2 has been indicated by colors. We expect that the stress expressed by Speech Modulation and Gesture Modulation (yellow) can be estimated from low level features, and therefore there is a direct mapping of these two variables across the two models. Speech Semantics Stress and Gesture Semantics Stress (purple) are more complex semantic variables, and therefore in the automatic model we propose to operationalize them by valence (ranging from positive to negative), arousal (ranging from passive to active) and a limited set of topics (content dependent clusters of words or gestures). Valence and arousal are measuring emotion in a continuous manner, so we expect that they can also be an indication of the stress level. The chosen speech and gesture topics are context dependent, and they contain semantic information on the amount of stress but also give qualitative insight into the type or cause of stress. The bar on the right side of the figure shows the stress level ranging from no stress (green) to very high stress (red).

In the next paragraphs we give a flavor of how stress prediction using our model with intermediate level variables works. More details about the intermediate level variables and how the ground truth was established for each of them are provided in Section 5, while the experimental setup including the low level features is described in Section 6.

The intermediate level variables are computed automatically based on human annotations. For Speech Modulation Stress and Gesture Modulation Stress we use the annotations from the testing of the human model. For speech valence and arousal, we use the ratings in the

ANEW database [30], for which many respondents were asked to rate words on a valence and arousal scale. We compute valence and arousal scores based on the spoken words and their occurrences and values in the ANEW list. We notice that the valence and arousal scores are based on human semantic interpretation which is an added value to the low level features. Another option would have been to compute the valence and arousal variables based on the acoustics of speech as was done in [31], but in our research we prefer to use the acoustics of speech for Speech Modulation Stress and the words for valence and arousal. The gesture valence and arousal are also based on human annotations. They are based on annotations of our recordings because an equivalent of ANEW for gestures is not available. The word topics and gesture topics are chosen based on key topics that are likely to appear during stressful interactions at a service desk, so they are context dependent. The word topics are computed based on appearances of specific words, and the gesture topics result from a human clustering of the gesture classes available in the service desk corpus.

The low level variables consist of features extracted from the audio and video stream. From speech we extract acoustic features, e.g. fundamental frequency (F0), intensity and voice quality features. In a fully automatic system, we would use a speech recognition system to receive the spoken words. For the purpose of this paper we assume perfect recognition and use manual transcription of the words instead. From the video stream we extract features related to movement and appearance. Ideally, there would be a symmetry between audio and video, and a gesture recognition unit would output a gesture type for which we would have valence and arousal scores as well as an associated meaning. Although gesture recognition is actively studied, to the best of our knowledge there is no gesture recognition module available to distinguish between the subtle and complex variations that appear when stress is conveyed. Therefore, we use the low level video features to predict the gesture-related intermediate level variables.

Below we give an account of the chosen variables that form the intermediate level:

- *Speech modulation stress*. The extent to which speech is conveying stress by how it sounds (e.g. prosody). This variable was annotated on a 3 point scale and automatically predicted from the acoustic features.
- *Speech valence*. How positive or negative the spoken words are. The value of this variable is computed based on the valence values of words in the ANEW list.
- *Speech arousal*. How active or passive the spoken words are. The value of this variable is computed based on the arousal values of words in the ANEW list.
- *Speech topics*. Based on keywords, we have created classes of words which have a relation to the level or the source of stress. Examples of speech topics are being helpless, late, aggressive and insulting.
- *Gesture modulation stress*. The extent to which the manner in which the gesture was done conveys

stress. This variable was annotated on a 3 point scale and automatically predicted from the video features.

- *Gesture valence.* We create a ground truth based on the valence annotation available for the database for gesture instances of each class. This variable is predicted from video features.
- *Gesture arousal.* We create a ground truth based on the arousal annotation available for the database for gesture instances of each class. This variable is predicted from video features.
- *Gesture topics.* We provide a clustering based on the 60 classes of gestures available in the dataset annotation. We define groups of gestures which have a semantic meaning with respect to stress which we call gesture topics. Examples of gesture topics are: explaining gestures, inner stress gestures, extrovert stress and aggressive. This variable is predicted from video features.

The idea behind this model was inspired by the large variability of manners in which people can express stress just by their voice and hand gestures, which makes the task of automatically assessing stress challenging. Furthermore, valence and arousal are entities used on many emotion related application and we expect that especially their combination correlates well with stress.

4 DATASET OF HUMAN-HUMAN INTERACTION AT A SERVICE-DESK

In order to test our stress models we validate them using our corpus of audio-visual recordings at a service desk, introduced in [32]. The dataset has been specifically designed for surveillance purposes. It contains improvised interactions of actors that only received roles and short scenario descriptions. The dataset is rich and includes various manifestations of stress. However, the actors did not receive any indication at all to encourage them to use their voice or body language in any particular way.

There have been debates on the use of acted corpora in emotion recognition related research. We argue that the use of actors is suitable for our purposes, due to the following considerations. Real-lived emotions are rare, very short, as well as constantly manipulated by people due to their desire to follow their strategic interests and social norms. Eliciting stress and negative emotions is not considered ethical. Furthermore, behavior of people in a real stressful situation will be influenced when they know that they are being recorded, which can also be considered as acting. In [33] it is being distinguished between push (physiologically driven) and pull (social regulation and strategic intention) factors of emotional expression. In interactions such as the ones from our scenarios at the service desk, we expect pull factors to have an important role. Since our aim is to study overt manifestations from the perspective of surveillance applications, we consider the use of actors appropriate. Even though actors are employed, the procedure was carefully designed to generate spontaneous interactions between actors.

In the remainder of this section we summarize the content, annotations and segmentations of the dataset, as they will be used in the experiments.

4.1 Content Description

The audio-visual recordings contain interactions improvised by eight actors. They had to play the roles of service desk employees and customers given only the role description and short instructions. Four scenarios were played two times, resulting in eight sessions, for which the actors did not see the performance of their colleagues from the other session. It resulted in realistic recordings, since the interaction between the actors built up as a result of the other's reactions which were not known beforehand. Example scenarios are a visitor who is late for a meeting and has to deal with a slow employee, a helpless visitor unable to find a location on a map asking the employee to be escorted but he is being refused, the service desk employee does not want to help because of his lunch break and the employee or a visitor is in a phone conversation and blocking the service desk.

Two cameras were used to record the interaction, and in our research we use the camera facing the visitor since we expect that most of the times the visitor is experiencing stress. However, it is sometimes the case that the employee is also visible in the camera facing the visitor. In cases when the employee is gesturing as well and that is visible, his/her gestures are also considered in our research. The total recordings span 32 minutes.

4.2 Annotations and Segmentation

The data as presented in [32] has annotations for the stress level by two annotators, utterance and gesture segmentations, speech transcriptions, as well as annotations of other dimensions such as valence and arousal of gestures, and classes of gestures. In addition to what was already available, in this paper we have added two more annotators for the stress level, resulting in an inter-rater agreement of 0.75 measured as Krippendorff's alpha. For segments that received different stress levels by the annotators, they had to check it and come together to a decision. Furthermore, we have annotated the degree of stress expressed by the four communication components proposed in Section 3.1 of this paper. These annotations were performed by one expert annotator and checked by a second one. Every time there would be a disagreement, the final ground truth would be set after discussion. For more details on the database and its annotations we refer to [32].

The utterance and gesture segmentations were done using the following procedure. As a basis, speech was segmented into utterances. First, borders were chosen based on turn taking. Whenever a turn was too long (i.e. longer than 10 seconds), it was split based on pauses. Separately, we have segmented the gestures that appear in the dataset. In general, we regard a gesture as one instance of a gesture class. A gesture class refers to gestures that have the same meaning and appearance. In the case of isolated gestures (i.e. not connected to other gestures or movements), the borders of the gestures are chosen by the annotators such that the segment includes the onset, apex and offset of the gesture. Gestures which include a repetitive movement, e.g. tipping of fingers were regarded as a whole, so there were no additional segments for each tipping movement. If a repetitive gesture was longer than 10 seconds, it was split

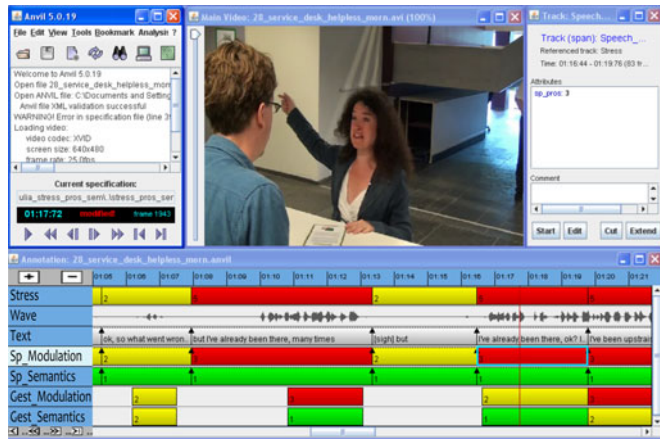


Fig. 3. Annotations for stress, speech semantics stress, speech modulation stress, gesture semantics stress and gesture modulation stress in Anvil [34].

into shorter segments. The two segmentations, and the annotation are outlined in Fig. 3. The figure also shows the annotation software Anvil [34] used by the raters.

The annotations used in this paper are:

- *Stress*. The perceived stress level in the scene was annotated on a five point scale based on multimodal assessment at the utterance level. In our experiments we simplify the problem to a three point scale, where classes 2 and 3 are treated as a class of moderate stress and classes 4 and 5 as high stress.
- *Speech semantics stress*. The extent to which stress is conveyed by the spoken words, annotated on a three point scale, at the utterance level.
- *Speech modulation stress*. The extent to which stress is conveyed by the way the message was spoken, annotated on a three point scale, at the utterance level.
- *Gesture semantics stress*. The extent to which a gesture conveys stress based on its meaning, annotated on a three point scale, at the gesture level.
- *Gesture modulation stress*. The extent to which a gesture conveys stress based on the way it is done, annotated on a three point scale, at gesture level.
- *Text transcriptions*. The text spoken by the actors has been transcribed. Besides the original words, several other sounds have been annotated: sighs, overlapping speech and rings from the bell available at the desk. The transcriptions are done at the utterance level.
- *Valence and arousal of gestures*. In [32] we identified 60 classes of gestures that have the same meaning and appearance. One representative instance of each class was annotated for valence and activation using the self assessment manikin [30]. The annotations of the selected gestures for each class were performed by eight raters using only the video (no sound) and another eight raters in a multimodal setup. In this work we use the average annotation of all 16 raters. The granularity for these annotations was a nine point scale, but we observed that almost all the data falls into the positive arousal and negative valence quadrant. To simplify the problem, we map to neutral the few labels for negative arousal and positive

valence. We end up with a five point scale for each dimension, which we reduce to a three point scale by coupling labels 2 and 3 together, and 4 and 5 together. The meaning of the new labels is 1—neutral arousal, neutral valence, 2—medium arousal, medium negative valence and label 3—high arousal and strongly negative valence.

- *Gesture class*. The 60 classes of gesture that have the same meaning and appearance found in [32] are used in this paper for gathering the Gesture Topics.

5 INTERMEDIATE LEVEL FEATURES

We continue with a description of the intermediate level variables from our model of automatic stress assessment, which were introduced in Section 3.2 and Fig. 2. We can divide them into two categories: variables for which we have an annotated ground truth and which we predict from low level visual and acoustic features (to be discussed in Section 6.1), and variables for which there is no ground truth available and for which we compute values based on simple algorithms. The prediction methodology is presented in Section 8.2. For both speech and gestures, given the complexity of semantic messages, the Semantic component was operationalized by three entities: Valence, Arousal and Topics.

5.1 Speech Modulation Stress and Gesture Modulation Stress

Speech Modulation Stress and Gesture Modulation Stress are the two variables which have been maintained from our human model. As a ground truth we use the dataset annotations described in Section 4.2. Speech Modulation Stress is predicted from the acoustic features, and Gesture Modulation Stress is predicted from the visual features.

5.2 Speech Valence and Speech Arousal

Speech Valence and Speech Arousal have no ground truths available. Our approach is to use a simple technique based on previous work for creating lists of words with valence and arousal scores. The Dictionary of Affect in Language (DAL) [35] and Affective Norms for English Words (ANEW) [30] the considered options. We use ANEW because it contains less but more emotion related words.

Since a significant part of the transcriptions is in Dutch, a first step was to use machine translation and obtain the English version. To maximize the matching between our words and the words from the ANEW list, we have applied stemming. The score of each utterance is initialized to neutral valence and arousal values. All the words from the utterance are looked up in the ANEW list. If matches are found, a new score is computed by averaging over the valence and arousal scores of the matching words. We find that 23 percent of the utterances contained words from the ANEW list, therefore the majority of the scores still indicated neutral valence and arousal.

5.3 Word Topics

Since we consider the service desk domain, we found that there are at least a number of topics which are likely to come up during visitor-employee interactions at the service desk which might be indications of stress. We defined five classes

of words indicating typical problems, and we have added the three nonverbal sounds which are annotated in the textual transcription. Together, they result in eight topics:

- *Being late*. Many times visitors at a service desk are stressed because they are late for a meeting. Keywords examples: 'late', 'wait', 'hurry', 'quickly', 'immediately'.
- *Helpless*. This indicated when visitors have difficulties finding what they need. Keywords include: 'no idea', 'need', 'help', 'difficult', 'find', 'problem'.
- *Dissatisfied*. Keywords examples: 'annoying', 'unkind', 'manager', 'rights', 'ridiculous', 'quit'.
- *Insults*. Keywords include curse and offensive words.
- *Aggressive*. Keywords examples: 'attack', 'sweat', 'guard', 'touched', 'hurts', 'push', 'police', 'control'.
- *Ring*. Annotated whenever someone would ring the bell to be found on the service desk.
- *Overlap*. Annotated when there was overlapping speech.
- *Sigh*. Sighs of the actors were also annotated.

Based on these topics, an utterance is described by an eight-dimensional feature vector containing counts of keywords from each topic. The number of occurrences of words from each topic can already give an impression of the content of the data. The most frequent topics are Late (20 percent of the utterances), Helpless (14 percent) and Dissatisfied (27 percent). Insults (2 percent) and Aggressive (4 percent) are less frequent. From the nonverbal indicators, the most frequent is Overlap (9 percent), followed by Sigh (3 percent) and Ring (2 percent).

5.4 Gesture Valence and Gesture Arousal

For Gesture Valence and Gesture Arousal we use the existing annotations for the representative gestures chosen from each class (recall Section 4.2). Even though valence and arousal of a gesture class might vary due to its modulation, we extend the label for one gesture class to all gestures in that class. The distribution of the resulting labels for arousal is 14 percent for class 1 (neutral), 61 percent for class 2 (medium) and 25 percent for class 3 (high arousal). For valence, the proportion of the middle class was even higher—69 percent, and the rest was formed from 19 percent class 1 (neutral) and 12 percent class 3 (very negative).

5.5 Gesture Topics

We want to recognize gestures for the purpose of stress prediction. Ideally, we would be able to recognize automatically all the gesture classes which were identified in the dataset. However, for many gesture classes there are very few or even just one example available. Taking into account our final application, identifying a smaller number of general categories of gestures might still be sufficient for stress prediction. Our approach was therefore to create a manual clustering of the 60 gesture classes into six classes, based on their meaning. We call each newly formed class a gesture topic. All the gestures which were assigned the same topic will have the same new label, and our problem transforms to recognizing to which topic a gesture belongs.

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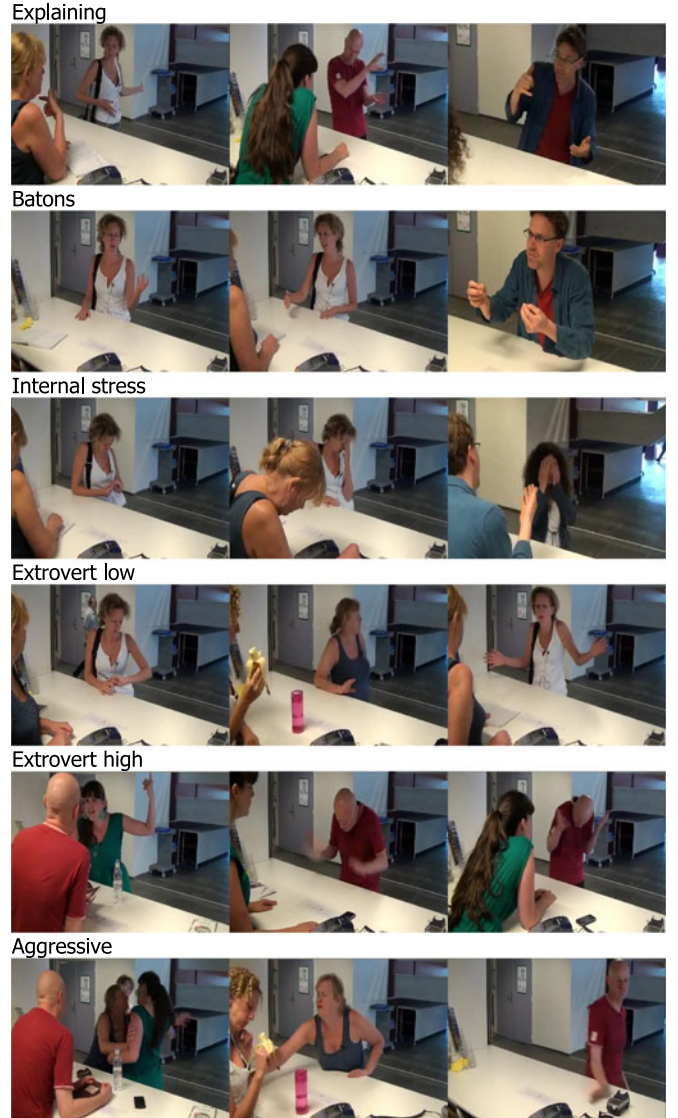


Fig. 4. Examples of gestures from each gesture topic.

The six topics are chosen by studying the application domain and the 60 classes that were already found. We define the following six topics, and examples from each of them are depicted in Fig. 4:

- *Explaining*. Gestures used to visually complement a linguistic description, explaining a concept, pointing directions, pointing to self and to others.
- *Batons*. As defined in [12] batons are gestures that accent or emphasize particular words or phrases. They are simple, repetitive, rhythmic movements with no clear relation to semantic content of speech.
- *Internal stress*. Gestures that point out to stress but without any signs of aggression or tendency to react. Example gestures are putting hand on the forehead, cleaning sweat, fidgeting, putting hands close to mouth, nose. This is the category in which self-adaptors (movements which often involve self touch, such as scratching) occur. Self adaptors are known to be indicative of stress. In [36] the effects of self adaptors as well as other language and gesture aspects on

perceived emotional stability and extroversion are studied.

- *Extrovert low.* Gestures that show stress and a tendency to make it visible to the opponent. E.g. tipping fingers or patting palm on the desk, pointing to watch.
- *Extrovert high.* Gestures that are clear indications of high stress, one step before becoming aggressive. For example slamming fists on the desk, wild movement with hands showing a lot of dissatisfaction.
- *Aggressive.* Gestures which clearly show aggression, like pushing somebody, slamming objects, throwing objects.

While we do not claim that these gesture topics provide a precise understanding of what a gesture means, we argue that an increase in stress probability can be observed from the first to the last gesture topic. We think these gesture topics offer a good coverage of the gesture types occurring in the database, and find them interesting candidates for stress prediction.

6 DATA PROCESSING APPROACH

This section provides details on the experimental setup. It contains a description of the low level features used for automatic stress assessment, as well as information about the segmentation and classification procedure which are relevant for both the automatic stress assessment and for the analysis based on annotations of how the speech and gesture components of our human model convey stress.

6.1 Low Level Feature Extraction

As described in Section 3.2, the acoustic and visual low level features are used to predict the intermediate variables and alternatively directly the stress level.

6.1.1 Acoustic Features

The acoustic features capture the nonverbal part of speech, namely, whether the manner in which somebody is speaking can convey signs of stress. Vocal manifestations of stress are dominated by negative emotions such as anger, fear, and stress. For the audio recognizers we use a set of prosodic features inspired from the minimum required feature set for emotion recognition proposed by [8] and the set proposed in [16]. The software tool Praat [37] was used to extract these features. Note that there are many more interesting features that can improve recognition accuracies. Example features that can be added are roughness, mode, MFCC, and spectral flux which can be extracted using the MIR toolbox [38], or feature sets from the INTERSPEECH paralinguistic challenge [39]. The goal of the paper is to explore the merit of an intermediate-level representation, and to assess this by verifying the relative improvement. Absolute performance can potentially be improved by adding more features, which is accommodated by the proposed framework and representation.

The hand crafted features set consists of the following features: speech duration without silences (for each utterance the speech part was separated from silences using an algorithm available in Praat [37]), pitch (mean, standard deviation, max, mean slope with and without octave jumps, and range), intensity (mean, standard deviation, max, slope

and range), first four formants (F1-F4) (mean and bandwidth), jitter, shimmer, high frequency energy (HF500) (HF1000), harmonics to noise ratio (HNR) (mean and standard deviation), Hammarberg index, spectrum (center of gravity, skewness), long term averaged spectrum (slope). As unit of analysis, we used the utterance segmentation.

6.1.2 Visual Features

We expect that the most relevant features for stress detection are based on movement. We chose to describe the video segments in terms of space-time interest points (STIP) [40], which are compact representations of the parts of scene which are in motion. Originally these features are employed for action recognition. They were successfully used in recognizing 48 actions in [41], while in [42] they were successfully applied for recognizing degrees of aggression. Employing more advanced features such as trajectory based features for genre detection can be included in the framework, but the purpose of this paper is to evaluate the use of the intermediate representation.

The space-time interest points are computed for a fixed set of multiple spatio-temporal scales. For the patch corresponding to each interest point, two types of descriptors are computed: histograms of oriented gradient (HOG) for appearance, and histograms of optical flow (HOF) for movement. For feature extraction we used the software provided by the authors [40].

These descriptors are used based on a bag-of-words approach, following [40]. We computed specialized codebooks, but instead of using K-means as in the original paper, the codebooks were computed in a supervised way using Random Forests with 30 trees and 32 nodes. We used the gesture segmentation provided in the dataset. The visual intermediate level features were predicted only for data for which gestures were available. The resulting feature vectors were reduced using correlation based feature subset selection.

6.2 Classification

All classification tasks were performed using a Bayes Net (BN) classifier. Each prediction task was treated as a set of binary one-against-all problems. The final prediction label was the one corresponding to the maximum posterior.

Due to the nature of human communication, and our utterance level and gesture level segmentations, there are data segments for which there are no gestures. These segments were handled by the Bayes Net classifier as missing data.

For almost all our prediction tasks there was an unbalance with respect to number of samples for each class. Therefore, we have tested the benefits of applying Synthetic Minority Over-sampling Technique (SMOTE) [43] to balance the data. The artificial samples were created for the training set and the proportion to be created was chosen such that the number of samples in the minority class should be at least 70 percent of the number of samples in the majority class.

As fusion methods, we experiment with feature and decision level fusion (DLF). Feature level fusion (FLF) means that the feature vector is formed by concatenating the low level acoustic and video features. In decision level fusion there are two stages of classification. In the first stage the

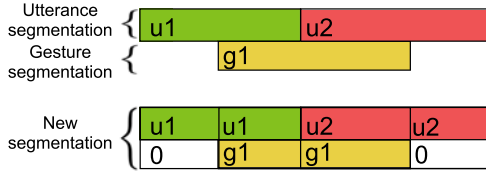


Fig. 5. The figure shows how a new segmentation was obtained based on the initial utterance and gesture segmentations.

acoustic and video low level features are used separately to predict a specific ground truth. In the second stage, the scores obtained from the first stage are used as features to predict the stress label.

All experiments were performed in a leave-one-session-out (LOSO) cross-validation framework. As performance measures, we report weighted (WA) and unweighted (UA) accuracies, to take into account data imbalance (recall Fig. 6). Confusion matrices are used to have a more detailed view on the performance of the classification task. Given the data unbalance and the fact that high recognition rates are desired for all classes, the unweighted average (UA) is considered the main performance measure.

6.3 Segmentation

The database contains two segmentations, one based on utterances, which was used for Stress and Speech Modulation Stress annotations, and one based on gestures. Many times the boundaries of utterances and gestures are not the same.

For prediction tasks involving both speech and gesture related variables, we adopt a new segmentation. For every utterance and gesture border, there is a border in the new segmentation, as shown in Fig. 5. The values of the variables are transferred based on timing. Take for example the case of two utterance with labels $u1$ and $u2$, and a gesture with label $g1$ starting within the first utterance and ending within the second utterance. In the new segmentation, there will be two utterances with label $u1$ with the middle border where the gesture began, two utterance with label $u2$ separated where the gesture ended, and two gestures both with label $g1$. This new segmentation was used only for the experiments that involved multimodal assessments of stress, and it was combined with a LOSO cross-validation scheme. In total 1,066 samples were used for classification, out of which 486 contained gestures.

7 ANALYSIS OF HOW SPEECH AND GESTURES CONVEY STRESS BASED ON THE HUMAN MODEL

In this section we explore how stress was expressed by the communication components defined in Section 3.1. The aim is to understand what are the dominant communication channels for conveying stress, which we expect to be also the most relevant in case of automatic prediction. For this task we use the labels described in Section 4.2 and the new segmentation explained in Section 6.3.

7.1 Analysis of the Correlations between the Communication Components and Stress

As a first step in our study, we are interested in correlation coefficients between the stress labels and the four

TABLE 1
Correlation Coefficients (Spearman) between Stress and the Proposed Communication Components. SSS = Speech Semantics Stress, SMS = Speech Modulation Stress, GSS = Gesture Semantics Stress and GMS = Gesture Modulation Stress

Stress	1				
SSS	.76	1			
SMS	.46	.53	1		
GSS	.52	.48	.25	1	
GMS	.15	.10	.10	.29	1
	Stress	SSS	SMS	GSS	GMS

annotated communication components. The annotated variables are ordinal and therefore we use Spearman correlation coefficients. Since part of the communication components are only gesture related and were annotated only when gestures appeared, we compute correlations for the variables which were available for all data, and separately for all variables only on the gesture segments. There were no significant changes in the correlation coefficients of the variables available for all data, if they were computed only on the gesture segments. Therefore, in Table 1 we show the correlations for all variables for the data segments that did contain gestures.

Stress is the most correlated with Speech Modulation Stress, followed by Speech Semantics Stress and Gesture Modulation Stress. A lower correlation coefficient is observed for Stress and Gesture Semantics Stress. Therefore, we expect the first three to be good measures for predicting stress. Gesture Semantics Stress was the least correlated with Stress. This implies that when gestures are used during stress, their modulation will give more indications of the emotional state than their semantics. Examples of gesture in this database that fall in this category are pointing gestures, which have semantically no relation to stress, but by their modulation can indicate stress. Apparently gestures conveying stress by their semantics, as for example insulting gestures, tapping or slamming hands on desk, or gestures with an aggressive tendency are less frequent.

We also computed in between communication components correlations. The two speech related components are highly correlated, as well as the SMS and GMS. From these values we expect that when the semantics of speech conveys more stress, that will be also noticeable from speech modulation. Also, when speech modulation is indicating more stress it is likely that it will be accompanied by gestures that also show more stress. One cause for the lower correlation between Stress and the gesture variables can be the fact that stress was assessed as a general measure of the scene, and not per person. This can lead to lower correlations when one actor is visible and the other one is speaking.

7.2 Characteristics of Speech and Gestures for Different Stress Levels

In this section we have a closer look on the use of the communication components per stress level. Fig. 6 presents three histograms and a number of gesture examples. Each histogram illustrates, given a specified stress level, the distribution of the annotations for the four stress communication components. The colors are associated with the four

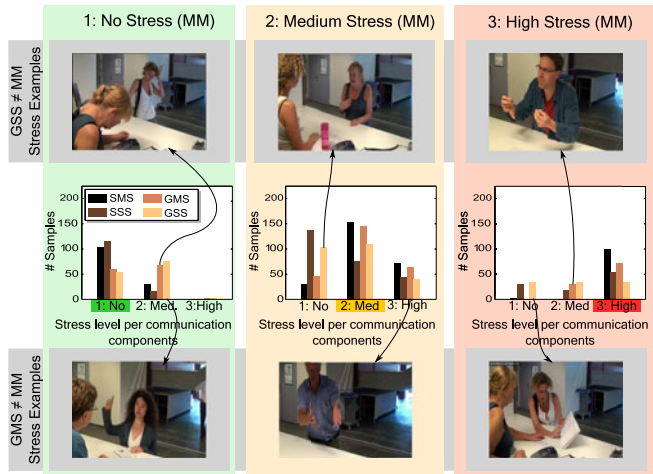


Fig. 6. The histograms represent the distribution of labels for Speech Modulation Stress, Speech Semantics Stress, Gesture Modulation Stress and Gesture Semantics Stress given a specified level of stress from the multimodal annotation (MM). We expect that the four communication components will indicate the same stress level perceived using multimodal data. The pictures represent example of when this is not the case, for either Gesture Semantics Stress (high) or Gesture Modulation Stress.

components, and the height of the bar represents the number of occurrences given the specified stress level. A missing bar means that there were no occurrences for that variable given the considered stress level.

Note that the three histograms were not normalized, and the numbers of occurrences on the vertical axis are applicable for all three of them. We do not normalize them because in this way they give an impression on how the annotations were distributed with respect to the stress level: 33 percent of the data fall into the no stress case, 49 percent into middle stress and 18 percent into high stress. With the gesture pictures we highlight interesting cases of gestures from different bins of the histograms. Their appurtenance to bins in the histograms is indicated by arrows.

The left histogram in Fig. 6 contains the distribution for when there was no stress. It can be observed that for these segments, Speech Modulation Stress and Speech Semantics Stress have predominantly label 1, so they also do not convey stress. Also, a significant amount of 62 percent of these segments did not contain any gestures. When gestures appeared, they were assigned label 1 or 2 for Gesture Modulation Stress and Gesture Semantics Stress. The most interesting situations are when they were labeled 2, since this is not expected for the no stress cases. Looking into examples from the data, we find that these gestures usually indicate little stress, like scratching head or keeping open palms on the desk. They do show the person is feeling a little tense, but from the overall conversation and given the context, the segment was not labeled as being stressful. Examples of such gestures are in the first column of gesture images from Fig. 6.

The middle histogram in Fig. 6 shows the distributions for medium stress. This time the proportion of segments which did not contain gestures decreased to 51 percent. Speech Semantics Stress is indicating no stress in 52 percent of the segments. This indicates that stress was perceived from other sources than words. Speech Modulation Stress is in this case the component that is most dominantly

indicating label 2, together with the two gesture components. Nevertheless, it can be observed that there are many combinations of the four components possible for which medium stress is assigned. This is a clear indication that automatically predicting the middle stress is challenging, and that is bound to generate confusion and make the recognition of the other stress classes difficult as well. The gesture pictures in the middle column of Fig. 6 are examples of gestures that appeared during medium stress, but which had label 1 or 3 for Gesture Modulation Stress and Gesture Semantics Stress.

For the high stress level, depicted in the rightmost histogram of Fig. 6, in 47 percent of the segments there were no gesture. What we conclude from this is a gradual increase in amount of gesticulation when stress increases. Out of the segments which did contain gestures, 70 percent indicate a high level of stress via modulation and 33 percent via their semantics. Speech Modulation Stress is dominantly indicating high stress in a proportion of 95 percent, and Speech Semantics Stress for 51 percent of the cases. Examples of gestures which indicated lower stress by their semantics and modulation but appeared in segments labeled with high stress are shown in the right column of Fig. 6.

What we learn globally from Fig. 6 is that the extreme cases are quite well indicated by the four variables: when there is no stress, or high stress, mostly the four communication components also indicate no and high stress, respectively. The most complex is the case of medium stress, for which many combinations are possible. The existence of all these possible combinations that lead to the same result, gives insight into the difficulty of fusing them for automatic stress prediction. For research dealing with how to fuse inconsistent information from audio and video in the context of multimodal aggression detection we refer to [44].

7.3 Predicting Stress from the Four Communication Components

As an experiment to see how well the labels of the four communication components can be used for predicting stress, we considered them as features and applied a Bayesian network classifier. The ground truth for the classifier was the stress label. This results in a 71 percent weighted average accuracy and 73 percent unweighted average accuracy when we consider all the data, and almost the same for when considering only the data for which gestures were visible. The confusion matrices for these two settings are shown in Table 2. As expected, due to the high variability observed in expressing stress, we can not achieve perfect accuracy even when we use the human labels for the four communication components. Note that even though the weighted average accuracies are equal for the two settings, for the gesture data setup the recall of class 3 which corresponds to the most stressful situations is significantly higher. This finding highlights the importance of gestures and also signifies the fact that performance in the all data setup suffers from missing data. These recognition results can be seen as the upper bound performance of the fully automatic stress assessment task.

To summarize, we observe from Table 1 that Speech Modulation Stress is a very good indication for stress. The

TABLE 2

Confusion Matrices in Percent for Predicting Stress from the Human Annotated Communication Components, for All Data (Left) and Only Gestures Data (Right)

	Classified as				Classified as		
	1	2	3		1	2	3
1	76	24	0	1	77	23	0
2	13	74	13	2	11	73	16
3	4	32	63	3	0	16	84
UA = 71				UA = 78			
WA = 73				WA = 76			

incidence of gestures increases gradually with the increase in stress level as seen in Fig. 6, which means that even the frequency of gestures can be an indication of stress. For no stress and high stress, the values of the four communication components consistently indicate the same stress level most of the times. The medium stress level is characterized by a variety of combinations of all values of the four components, making it more difficult to come to a conclusion. Finally, when using the labels of the four communication components to predict stress we achieve a UA of 71 percent for all data, and of 78 percent if we consider only the segments that contain gestures.

8 AUTOMATIC STRESS ASSESSMENT—RESULTS AND DISCUSSION

This section gives insight into the results for stress prediction using the model proposed in Fig. 2, and compare it to a baseline of predicting stress directly from low level features. The section is organized in three parts: Section 8.1 gives results for the baseline method, Section 8.2 focuses on automatic prediction of the intermediate level variables from our automatic model (recall Fig. 2), and Section 8.3 presents the final results for automatic stress assessment using our model.

8.1 Baseline: Predicting Stress from Low Level Features

The baseline results are for predicting stress directly from low level features. We provide results for using the acoustic features only, the video features only, as well as for feature and decision level fusion. Table 3 shows classification results for the BN classifier, given LOSO cross-validation and the segmentation described in Section 6.3. It contains accuracies per class as well as their weighted and unweighted average.

From the results in Table 3 we notice that the audio features were better predictors than the video ones, and that

TABLE 3
Baseline: Predicting Stress from Low Level Features (Bayesian Classifier)

Features	Fusion	UA	WA
acoustic	-	64	62
text	-	47	51
STIP	-	29	34
both	FLF	61	62
both	DLF	62	61

TABLE 4

Confusion Matrices in Percent for Automatically Predicting Stress Using Decision Level Fusion for the Baseline Approach

	Classified as		
	1	2	3
1	79	20	0
2	31	57	13
3	6	47	47
UA = 62			
WA = 61			

both feature and decision level improve over acoustic only. In Table 4 we show the confusion matrix for decision level fusion, which provides a good balance between UA and WA, as well as high recognition rates.

We continue with the setup and results for predicting the intermediate level variables and the stress label using the model we proposed.

8.2 Automatic Prediction of the Intermediate Level Variables

For Speech Modulation Stress, the unit of analysis was the utterance segmentation. For the four gesture related variables, the unit of analysis was the gesture segmentation. This means that while for Speech Modulation Stress we did analyze the whole data, when learning the four gesture related variables we used only the part of the data which contained gestures.

For each prediction task we tested the three classifiers mentioned above, and experimented with applying or not the SMOTE technique to deal with data imbalance. Because predicting these variables is an intermediate step that affects the final stress prediction, we use the best performing approach in each case, as indicated in Table 5.

Table 5 shows that Speech Modulation Stress is predicted with the highest accuracy. Gesture Valence is also predicted with high accuracy, followed by Gesture Modulation Stress. Given that the same generic low level features are used for all these task, we consider the results satisfying. Furthermore, the results might have been affected by the high degree of approximation in the labels of Gesture Valence, Gesture Arousal and Gesture Topics, since they are generalized from labels of only one instance of each gesture class.

The Gesture Topics variable is the only one for which instead of a three class problem, there was a six class problem. Given that predicting these intermediate variables automatically from low level features were three or six class

TABLE 5
Weighted (WA) and Unweighted (UA) Accuracies for Predicting the Intermediate Level Variables from Low Level Features

Features	Predicted variable	SMOTE	UA	WA
acoustic	Speech Modulation Stress	0	66	64
STIP	Gesture Valence	1	68	73
	Gesture Arousal	1	59	64
	Gesture Topics	0	57	61
	Gesture Modulation Stress	0	68	65

TABLE 6
Predicting Stress Based on the Intermediate Level Variables
(Bayesian Classifier), No SMOTE, for all data

Features	Fusion	UA	WA
SMS	-	66	62
SMS & text	DLF	63	62
SMS & GMS	DLF	64	64
SMS & GV	DLF	67	64
SMS & GA	DLF	65	62
SMS & GT	DLF	67	64
all	DLF	62	62
all selected	DLF	69	66

SMS = Speech Modulation Stress, SV = Speech Valence, SA = Speech Arousal, ST = Speech Topics, GMS = Gesture Modulation Stress, GA = Gesture Arousal, GV = Gesture Valence and GT = Gesture Topics.

problems, three or six respectively soft decision values (posteriors) are passed to the feature set for the final stress assessment. The intermediate level variables are predicted with different accuracies, which can also have an effect on how valuable the features are for stress prediction.

8.3 Predicting Stress Using Intermediate Level Variables

Table 6 presents results for stress prediction using our model with intermediate level variables. We study the performance of each intermediate variable independently. Starting with Speech Modulation Stress, the best performing variable, we search for the best combination of two variables. Finally, we show the results obtained when all intermediate level variables are used.

When studying the performance of using each single feature type in turn, we observe that Speech Modulation Stress is performing the best. A problem that appears with the other words-related or gesture-related features is their sparsity. For the words-related features, a problem is that the words from the ANEW list do not appear frequently in the spontaneous speech of the actors. Also, the keywords corresponding to the Speech Topics do not have a high frequency. Most information conveyed linguistically regards information unrelated to the emotional state of the speaker, such as explaining directions or a situation. The same problem occurs for gestures, since they are available for only part of the data, and treated as missing data by the BN when not present.

The weighted and unweighted average accuracies of fusing Speech Modulation Stress with each of the other intermediate level variables are shown in Table 6. It can be noticed

TABLE 7
Confusion Matrices in Percent for Predicting Stress Using
Speech Modulation Stress for All Data (Left), and Our Approach
with Feature Selection for All Data (Right)

	Classified as				Classified as		
	1	2	3		1	2	3
1	77	21	2	1	80	18	2
2	33	49	17	2	31	53	16
3	5	25	70	3	6	19	74
	UA = 66				UA = 69		
	WA = 62				WA = 66		

TABLE 8
Confusion Matrix in Percent for Predicting Stress Given Our
Approach, with Feature Selection Only on Gesture Segments

	Classified as		
	1	2	3
1	74	24	2
2	25	60	15
3	4	23	73
	UA = 69		
	WA = 67		

that adding almost any other variable does not cause significant changes to the result. The best performance is achieved by fusing Speech Modulation Stress with the Gesture Topics and with Gesture Valence. Another interesting phenomenon is that adding more features does not always improve performance, and using all features is slightly worse than using only the Speech Modulation posteriors. This is probably because the classifier is fed less relevant features compared to the first 3 very good features, and the performance drops. However, by running feature selection on this final feature vector, we obtain the best result. The selected feature set consists of the following 10 features: the three posteriors of Speech Modulation, text arousal, the aggressive and sighs speech topics, the posteriors of gesture classes extrovert low and extrovert high. We observe a consistent increase in performance of on average 4 percent for all stress levels using these features in addition to the SMS posteriors.

The confusion matrices for using only Speech Modulation and the selected features are shown in Table 7, left and right respectively, while results obtained from an experiment only on the segments that contain gestures and feature selection is presented in the confusion matrix in Table 8.

By inspecting which samples from the data were well classified by Speech Modulation Stress only and which ones benefited by adding gestures, we noticed a number of interesting cases. For example, it can be the case that the employee is speaking in a calm manner, but the visitor is visible and his gestures are indicating stress. It can also be the case that there is no speech (this happens rarely in our data and only for very short time intervals), and in that case gestures are the only indication we get. This situation also appears when there is a case of physical aggression, e.g. throwing an object or pushing, which are sudden movements and sometimes not accompanied by any sounds. All these cases would have been missed without using gestures.

It is interesting that Gesture Topics and Gesture Valence are performing best in combination with Speech Modulation Stress. Gesture Topics has the property of indicating a degree of stress (the topics can be ordered with respect to stress), which can be seen as a quantitative function. Besides, it can also be seen as a qualitative indicator of stress, since the topics have this property and especially discriminate between stress types. Gesture Valence is an indicator of how positive or negative a gesture is, and therefore has a direct relation with the degree of stress.

All in all, the performance achieved using our approach significantly improves over the baseline. When comparing the per class accuracies of our approach with feature selection, (presented in Table 7(right)), compared to standard decision level fusion (Table 4), we observe a dramatic

increase of 27 percent in the recall of class 3, high stress, in the detriment of 3 percentages in class 2 (medium stress). Given our envisioned application in the surveillance domain, this significant improvement for recognizing high stress is very beneficial since we are particularly interested in not missing samples of medium and high stress.

When comparing the results achieved by using the human labels of the four communication components from Table 2 (left), to the results achieved by automatic prediction using our final approach (confusion matrix in Table 7(right)), we notice that the automatic prediction yields better performance for high stress. Furthermore, the unweighted average accuracy for automatic stress prediction is only 2 percent absolute lower than the prediction based on the human labels, and the unweighted average is 7 percent lower.

9 SUMMARY AND CONCLUSION

To summarize, in the framework of automatic surveillance, we investigated modalities of how speech and gestures communicate stress and how they can be used for automatic assessment of stress. For this purpose we proposed a human model of stress communication, which distinguishes between the semantics expressed by speech and gestures, and the way in which the messages are delivered (modulation). We assessed how these components convey stress based on human annotated labels. As a next step, we proposed a new method for automatic stress prediction based on a decomposition of stress into a set of intermediate level variables. The intermediate level variables were obtained by operationalizing the communication components of the human model. We validated our model for automatic stress prediction and obtained significant improvements over a baseline predictor based on decision level fusion on the audio, text and video features.

To conclude, we answer the four research questions stated in the introduction. The first research question concerns the contribution of verbal and nonverbal communicative acts in conveying stress. Based on the analysis performed on our human model, we suggest that Speech Semantics Stress and Gesture Semantics Stress are considered verbal communication, while Speech Modulation Stress and Gesture Modulation Stress as nonverbal communication. Our findings point out that nonverbal communication, and in particular Speech Modulation Stress is the most dominant in communicating stress. However, we learned that stress is conveyed by a large variety of combination of these communicative acts, and if not considering them all we might miss the correct scene interpretation.

The second question refers to the contribution of gestures in stress communication and stress assessment. From the human model study we found that especially Gesture Modulation Stress is highly correlated with stress level. Furthermore, we observed an increase in the gesture frequency as the stress becomes higher. When automatically assessing stress based on a single intermediate feature type, Speech Modulation Stress had the best performance. When evaluating combinations of two intermediate level features, it combined best with Gesture Topics. In general adding gesture information did not lead to high improvements. However, by examining the samples for which they had a positive

impact we found that these were difficult cases, mostly from the middle and high stress categories. Examples are: stressful gestures of the visitor accompanied by calm speech of the employee, or stressful gestures without any speech.

The third question relates to the choice and performance of the intermediate level features. The best performing feature was Speech Modulation Stress, and it combined best with the Gesture Topics. However, it must be noted that the performance of a number of other variables was negatively influenced by their sparsity.

Finally, to answer the fourth question, we state that our method for stress prediction based on intermediate level variables significantly improves over the baseline of predicting stress from low level audio-visual features. Furthermore, the increase in performance is dramatic in the high stress class, which is highly beneficial for the envisioned application.

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Iulia Lefter is the corresponding author.

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